

HUALIEN AP, Taiwan

WMO: 467630

Lat: 24.023N

Lon: 121.618E

Elev: 16

StdP: 101.14

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.7	12.8	7.8	6.6	13.5	8.9	7.1	14.0	11.4	16.6	10.3	16.7	2.4	240	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	33.0	27.8	32.2	27.5	31.9	27.4	28.8	31.7	28.3	31.3	27.9	31.0	4.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.2	31.3	27.2	23.1	30.7	27.0	22.7	30.5	93.9	31.8	91.5	31.8	89.7	31.0	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.7	7.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			7.8	30.0	1.6	1.1	6.6	30.8	5.7	31.5	4.8	32.1	3.7	32.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.4	18.0	18.0	19.9	22.7	25.1	27.1	28.5	28.5	26.8	24.6	21.9	19.3		
	DBStd	4.32	2.32	2.67	3.08	2.69	2.01	1.68	1.08	1.11	1.56	1.64	2.17	2.29		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	111	34	36	20	2	0	0	0	0	0	0	3	17		
	CDD10.0	4883	248	224	307	382	468	512	573	503	452	356	287	27		
	CDD18.3	1953	23	26	69	134	209	262	314	314	253	193	108	46		
	CDH23.3	16692	42	59	236	736	1551	2692	3773	3691	2336	1138	349	91		
	CDH26.7	5423	1	2	23	142	372	921	1537	1491	681	205	43	5		
(14) Wind	WSAvg	2.7	3.0	2.8	2.7	2.5	2.4	2.4	2.3	2.4	2.7	3.1	3.3	3.1		
(15) Precipitation	PrecAvg	2129	67	101	88	111	204	229	179	236	353	357	133	54		
	PrecMax	3558	200	338	290	369	600	698	791	679	966	1700	428	143		
	PrecMin	989	12	24	24	26	30	26	14	3	32	12	4	15		
	PrecStd	556	36	77	50	78	122	157	163	188	238	377	112	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	26.8	28.8	31.1	31.6	32.9	33.7	34.1	32.2	31.2	29.1	27.1		
		MCWB	22.1	22.8	24.3	26.1	27.1	27.7	28.2	28.4	27.6	26.3	25.3	22.8		
	2%	DB	24.4	25.1	27.1	29.3	30.6	32.0	33.0	33.1	31.8	29.9	28.0	25.3		
		MCWB	20.7	21.5	23.4	25.0	26.5	27.3	27.8	27.8	27.3	25.7	24.2	21.5		
	5%	DB	23.1	23.9	26.0	28.2	29.8	31.2	32.2	32.2	31.0	28.9	26.8	24.2		
		MCWB	19.9	20.7	22.9	24.6	26.1	27.2	27.6	27.5	27.1	25.1	23.2	20.6		
	10%	DB	21.9	22.2	24.8	27.1	28.9	30.8	31.9	31.9	30.1	27.8	25.2	23.1		
		MCWB	19.1	19.6	22.1	23.9	25.5	27.1	27.5	27.4	26.6	24.5	22.3	19.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.4	23.2	25.2	26.6	28.0	28.9	29.4	29.5	28.8	27.3	26.0	23.5		
		MCDB	25.1	25.8	27.8	29.8	30.6	31.3	32.1	32.3	31.4	29.6	28.3	26.2		
	2%	WB	21.2	22.0	24.1	25.8	26.9	28.1	28.8	28.7	28.0	26.4	24.9	22.1		
		MCDB	23.7	24.4	26.6	28.8	29.6	30.8	31.8	31.9	30.8	28.8	27.4	24.7		
	5%	WB	20.4	21.0	23.1	25.0	26.3	27.7	28.2	28.2	27.2	25.7	23.9	21.1		
		MCDB	22.7	23.3	25.4	27.9	28.9	30.4	31.4	31.4	30.1	28.0	25.9	23.5		
	10%	WB	19.4	20.1	22.1	24.2	25.8	27.1	27.8	27.7	26.7	24.9	22.9	20.3		
		MCDB	21.5	22.1	24.3	26.7	28.3	30.0	31.0	31.0	29.4	27.0	24.7	22.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	6.0	6.2	6.3	6.2	6.6	6.9	7.0	6.4	6.3	6.2	6.5		
		MCDBR	8.6	8.5	8.0	7.7	7.2	7.2	7.4	7.6	7.4	7.7	8.0	8.3		
	5% WB	MCWBR	6.2	6.3	6.0	5.6	5.1	5.0	5.0	4.9	4.9	5.0	5.4	5.9		
		MCWBR	8.0	8.0	7.7	7.3	7.0	6.8	7.1	7.2	6.9	7.0	7.2	7.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.541	0.576	0.697	0.703	0.613	0.551	0.528	0.561	0.559	0.543	0.532	0.547		
		taud	1.899	1.857	1.623	1.662	1.886	2.054	2.124	2.013	2.004	1.998	2.001	1.914		
	Ebn at Noon	Ebn at Noon	705	715	652	662	722	762	781	755	747	734	711	680		
		Edh at Noon	177	196	260	254	202	170	158	176	175	168	158	169		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.38	2.78	3.16	3.52	4.12	4.90	5.72	5.14	4.51	3.75	2.76	2.36		
	RadStd	RadStd	0.34	0.45	0.46	0.55	0.68	0.58	0.36	0.51	0.48	0.40	0.29	0.36		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	+0.27	N/A	N/A	N/A	N/A	N/A	+81

Nomenclature: See separate page